DPIRD Field Mapping Tool

User Documentation

Linux / macOS Edition

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1. Overview

The DPIRD Field Mapping Tool is a desktop application designed for digitising and managing agricultural field trial plots. It runs locally as a web application in your browser and provides four core functions:

- **Generate Grid** — Create regular plot grids over drone orthomosaic imagery.
- **Edit Grid** — Interactively edit plot polygons in a browser-based map editor with tools for dragging, vertex editing, multi-select, copy/paste, and real-time measurements.
- **Convert File** — Convert between Shapefile, GeoJSON, GeoPackage, KML, and HDF5 formats with optional coordinate system reprojection.
- **Cropping Tool** — Crop raster data (orthophotos, DSMs) to individual plot polygon boundaries.

The tool requires no internet connection after installation and processes all data locally on your computer.

2. Installation

This guide covers installation on Linux and macOS. Users are expected to have Anaconda or Miniconda already installed.

2.1 Prerequisites

- Linux (Ubuntu 20.04+, Fedora, etc.) or macOS (11 Big Sur or later)
- Anaconda or Miniconda installed and available on your PATH
- Internet connection (first-time setup only)
- Approximately 2 GB disk space

If you do not have conda installed, download Miniconda from:
https://docs.anaconda.com/miniconda/

2.2 Installation Steps

1. Extract the provided zip file to any location on your system.

```
unzip DPIRD_Field_Mapping_Tool.zip -d ~/Tools/
cd ~/Tools/DPIRD_Field_Mapping_Tool
```

2. Make the shell scripts executable.

```
chmod +x install_dpird_tool.sh run_dpird_tool.sh
```

3. Run the installer.

```
./install_dpird_tool.sh
```

The installer will:

- Create a conda environment named dpird_field_tool with Python 3.11
- Install geospatial packages (numpy, pandas, geopandas, rasterio, fiona, shapely, pyproj, pillow, matplotlib, psutil) from conda-forge
- Install Streamlit and web packages (streamlit, streamlit-folium, folium, leafmap, localtileserver, scooby) via pip
- Verify that all packages imported correctly

This process takes approximately 10–15 minutes depending on your internet speed. You only need to do this once.

2.3 Manual Installation (Alternative)

If you prefer to set up the environment yourself:

```
# Create and activate the environment
conda create -n dpird_field_tool python=3.11 -y
conda activate dpird_field_tool

# Install geospatial stack from conda-forge
conda install -c conda-forge -y \
    numpy pandas geopandas rasterio fiona shapely pyproj \
    pillow matplotlib psutil

# Install Streamlit and web packages
pip install streamlit streamlit-folium folium leafmap \
    localtileserver scooby
```

Verify the installation:

```
python -c "import
numpy,pandas,PIL,shapely,pyproj,fiona,rasterio,geopandas,matplotlib,psutil,foliu
m,streamlit,streamlit_folium,leafmap"
```

If this runs without errors, installation is complete.

2.4 Running the Tool

Run the launch script:

```
./run_dpird_tool.sh
```

The script will activate the conda environment, launch Streamlit, and open your default browser at http://localhost:8501. If the browser does not open automatically, navigate to that address manually.

Alternatively, launch manually:

```
conda activate dpird_field_tool
cd /path/to/DPIRD_Field_Mapping_Tool
streamlit run app/field_mapping_tool.py \
    --server.headless true \
    --browser.gatherUsageStats false \
    --server.port 8501
```

2.5 Stopping the Tool

You can stop the tool by either:

- Pressing **Ctrl+C** in the terminal where the tool is running.
- Clicking the **“Close Tool”** button in the sidebar of the application.

2.6 Uninstalling

Remove the conda environment and delete the tool folder:

```
conda env remove -n dpird_field_tool
rm -rf /path/to/DPIRD_Field_Mapping_Tool
```

3. Interface Overview

The application has four tabs across the top and a sidebar on the left.

3.1 Sidebar

- **Logo** — Displays the organisation logo (if present in the bin/ folder).
- **Map Settings** — Toggle to show or hide the background basemap layer (CartoDB tiles).
- **How to Use** — Quick reference for each tool tab.
- **Documentation** — Expandable sections with detailed parameter descriptions.
- **Close Tool** — Shuts down the application.

3.2 Tabs

The main area contains four tabs. Click a tab name to switch between tools. Each tab operates independently.

4. Generate Grid

This tab creates a regular grid of plot polygons over an orthomosaic image within a user-drawn boundary.

4.1 Workflow

4. Enter your project folder path in the text field, or click Browse to select it. The folder should contain your orthomosaic (.tif) and an optional CSV file.
5. Select the orthomosaic file from the dropdown. Optionally select a CSV file to attach additional plot metadata.
6. Set the grid parameters: Banks, Rows, Buffer, and Plot Size.
7. Draw a boundary polygon on the map by clicking the four corners of the trial area.
8. Click Generate Grid.
9. Review the generated grid on the map and in the data table.
10. Use the action buttons to Edit, Save the grid, or Save the boundary.

4.2 Parameters

Parameter	Description
Banks	Number of banks (columns) in the X direction. Each bank contains a vertical stack of rows.
Rows	Number of rows in the Y direction within each bank.
Buffer (m)	Gap between adjacent plots in metres. A value of 0.1 leaves a small gap; 0 means plots touch edge-to-edge.
Plot Size (W,H)	Width and height of each plot in metres, comma-separated. For example, 4,1 creates plots 4 m wide by 1 m tall.

4.3 Boundary Drawing Order

Draw the four corners in this order:

Top-left (B1R1) → Top-right (BXR1) → Bottom-right (BXRY) → Bottom-left (B1RY)

B refers to Bank and R refers to Row. The first point you click will be adjacent to plot B1R1 (the top-left corner of the trial design).

4.4 Plot ID Convention

Field	Formula	Examples
Plot_ID	Bank × 1000 + Row	B1R1=1001, B1R2=1002, B2R1=2001, B3R4=3004
B/R	Bank-Row label	B1R1, B2R3, B12R6
Bank	Bank number	1, 2, 3, ...
Row	Row number	1, 2, 3, ...

4.5 CSV Input (Optional)

The CSV file is entirely optional. If provided, its columns are merged as additional metadata into the plot attributes based on plot order. Without a CSV, the tool automatically generates Plot_ID, B/R, Bank, and Row fields from the boundary vertices and grid parameters.

4.6 Action Buttons

Button	Action
Generate Grid	Creates the grid from the drawn boundary and current parameters.
Clear Grid	Removes the current grid and boundary so you can start over.
Edit Grid	Opens the interactive browser-based polygon editor in a new tab.
Save Initial Grid	Saves the generated grid as a shapefile in the project folder.
Save Trial Boundary	Saves the drawn boundary polygon as a shapefile.

5. Edit Grid

This tab allows you to load an existing grid shapefile and orthomosaic, then open an interactive browser-based map editor to modify the plot polygons.

5.1 Workflow

11. Click Load Grid Shapefile and select your grid file.
12. Click Load Orthomosaic and select the TIF file.
13. Click Create & Open Editable Map to launch the browser editor.
14. Use the editing tools (described below) to modify polygons.
15. Click Save Shapefile in the browser editor to write changes to disk.

5.2 Browser Editor Tools

Tool	Shortcut	Description
Navigate	Esc	Pan and zoom the map. Default mode.
Drag Plots	D	Click and drag polygons to reposition. Ctrl+Click to multi-select, then drag all selected together.
Edit Vertices	V	Drag individual corner vertices to reshape polygons. If polygons are selected, only those are editable.

Delete	X	Click a polygon to remove it. If multiple are selected, all selected are deleted.
Draw New	N	Click to place vertices, double-click or click first vertex to close the polygon.
Measurements	M	Toggle edge length labels in metres on all polygons. Updates in real time during vertex editing.
Copy	Ctrl+C	Copy the selected polygon to clipboard.
Paste	Ctrl+V	Paste a duplicate polygon nearby with a new Plot_ID.
Apply Shape to Selected	—	Copies the first selected polygon's shape to all other selected polygons, centered on each centroid.
Apply Shape to All	—	Copies the selected polygon's shape to every other polygon on the map.
Undo	Ctrl+Z	Revert the last editing action. Up to 50 undo steps.
Select All	Ctrl+A	Select all polygons on the map.

Note: On macOS, use **Cmd** instead of **Ctrl** for all keyboard shortcuts (Cmd+Click, Cmd+C, Cmd+V, Cmd+Z, Cmd+A).

5.3 Selection

- Click a polygon to select it (highlighted in blue).
- Ctrl+Click (Cmd+Click on macOS) or Shift+Click to add or remove polygons from the selection.
- Ctrl+A (Cmd+A) selects all. Esc clears the selection.
- The status bar at the bottom shows the current mode, total plots, and selected count.

5.4 Saving

- **Save Shapefile** — Writes the edited grid directly to the project folder as a shapefile.
- **Export GeoJSON** — Downloads a GeoJSON file to your browser's downloads folder.

6. Convert File

This tab converts vector geospatial files between formats with optional coordinate reference system (CRS) reprojection.

6.1 Supported Formats

Input Formats	Output Formats
Shapefile (.shp), GeoJSON (.geojson/.json), GeoPackage (.gpkg), KML (.kml), HDF5 (.h5/.hdf5)	Shapefile (.shp), GeoJSON (.geojson), GeoPackage (.gpkg), KML (.kml)

6.2 CRS Options

Option	Description
Source file CRS	No reprojection. Keeps the original coordinate system.

WGS 84 (EPSG:4326)	Global geographic coordinates (latitude/longitude).
GDA2020 / MGA Zones 49–56	Australian map grid zones on the current datum.
GDA94 / MGA Zones 49–56	Australian map grid zones on the older datum.
Custom EPSG	Enter any EPSG code manually (e.g. EPSG:32750).

Note: KML output is always reprojected to WGS 84 as required by the KML format specification.

6.3 Workflow

16. Click Browse Input File and select the vector file to convert.
17. Choose the output format from the dropdown.
18. Select the target coordinate system (or enter a custom EPSG code).
19. Click Choose Save Folder and select where to save the output.
20. Enter an output filename (without extension).
21. Click Convert & Save.

7. Cropping Tool

This tab crops raster data (orthophotos, DSMs, or any GeoTIFF) using polygon boundaries.

7.1 Two Modes

Single polygon (e.g. a trial boundary): Saves one cropped raster with the filename you specify.

Multiple polygons (e.g. a plot grid): Saves one raster per polygon, named filename_PlotID.tif (e.g. cropped_1001.tif, cropped_2003.tif).

7.2 Workflow

22. Click Load Shapefile / Vector and select a boundary file (Shapefile, GeoJSON, or GeoPackage).
23. Click Load Raster (TIF) and select the raster file to crop.
24. Click Choose Save Folder and select where to save the cropped output.
25. Enter a base filename (without extension).
26. Click Crop and Save.

The tool automatically reprojects the vector to match the raster's CRS if they differ. Pixels outside the polygon boundary are set to the raster's nodata value (or 255 for uint8 imagery) to preserve valid dark pixels within the plots.

8. Map Settings

The sidebar contains a Basemap toggle under Map Settings.

- **OFF (default):** Maps display only your orthomosaic and polygons on a blank background. This avoids "Access blocked" errors from tile servers.
- **ON:** A CartoDB Positron basemap is shown underneath your data, providing geographic context such as roads and place names.

Your orthomosaic covers the trial area completely, so the basemap is typically not needed and can be left off.

9. Platform-Specific Notes

9.1 File/Folder Browser Dialogs

The tool uses tkinter file dialogs for browsing files and folders. On some Linux distributions, tkinter may not be installed by default. If the Browse buttons do not work:

Ubuntu / Debian:

```
sudo apt-get install python3-tk
```

Fedora / RHEL:

```
sudo dnf install python3-tkinter
```

macOS: tkinter is included with the Python from conda, so no additional installation is needed.

Workaround: If tkinter dialogs do not appear or open behind other windows, you can type or paste file/folder paths directly into the text input fields in the application. The Browse buttons are a convenience, not a requirement.

9.2 Display Server (Linux)

The tkinter file dialogs require a display server (X11 or Wayland). If you are running on a headless server or via SSH without X-forwarding, the Browse buttons will not work. Use the text input fields to enter paths directly instead.

9.3 Port Conflicts

If port 8501 is already in use, the launch script will attempt to kill the stale process. You can also specify a different port manually:

```
conda activate dpird_field_tool
streamlit run app/field_mapping_tool.py --server.port 8502
```

9.4 macOS Keyboard Shortcuts

On macOS, use **Cmd** instead of **Ctrl** for all editor shortcuts: Cmd+Click (multi-select), Cmd+C (copy), Cmd+V (paste), Cmd+Z (undo), Cmd+A (select all).

10. Troubleshooting

Tool shows a white screen

A previous session may still be running on port 8501. Press Ctrl+C in the terminal to stop it, then relaunch. Or kill it manually:

```
lsof -i :8501 -t | xargs kill -9
```

Browse button opens but nothing happens

A file picker dialog may have opened behind the browser window. Check your taskbar or dock. Alternatively, type or paste the file path directly into the text input field. See Section 9.1 for tkinter installation instructions.

Access blocked message on the map

This occurs when the basemap tile server blocks requests from localhost. Turn off the basemap toggle in the sidebar — your orthomosaic provides full coverage of the trial area.

Installation fails at package install step

Ensure you have an active internet connection. If individual packages fail, re-run install_dpird_tool.sh — it will detect missing packages and retry. For persistent issues with rasterio or fiona, these packages require C libraries (GDAL, PROJ) which conda-forge provides reliably. Avoid installing these via pip on Linux unless you have the system libraries pre-installed.

Raster crop output has white pixel holes

This was a known issue in earlier versions where nodata was set to 0, causing valid black pixels to appear as holes. The current version uses a safe nodata value (255 for uint8, NaN for float) that preserves dark pixels within plots.

"conda not found" error

Ensure Anaconda or Miniconda is installed and conda init has been run for your shell. Restart your terminal after installation. You can verify with:

```
conda --version
```

Permission denied on scripts

Run the following to make the scripts executable:

```
chmod +x install_dpird_tool.sh run_dpird_tool.sh
```

11. Included Files

File	Purpose
install_dpird_tool.sh	One-time setup: creates conda environment and installs all packages.
run_dpird_tool.sh	Launches the tool (activates env, starts Streamlit, opens browser).
app/field_mapping_tool.py	Main application code.
bin/logo.png	Organisation logo displayed in the sidebar (optional).
INSTALL_LINUX_MACOS.md	Quick-reference installation guide in Markdown format.
README.txt	Quick-start reference.

12. Author & Acknowledgements

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